

Joint meromorphic boundaries of two-clock common returns

Abstract

For integers $a, b \geq 2$, we study the ordinary two-variable series whose coefficient at $x^n y^m$ is the number of common fixed points of the n -th iterate of circle multiplication by a and the m -th iterate of circle multiplication by b . These coefficients are $\gcd(a^n - 1, b^m - 1)$. We classify the open domain of absolute convergence and the joint meromorphic continuation for every pair of bases. For multiplicatively independent bases the convergence domain is the unit bidisc. For $a = c^r, b = c^s$ with coprime positive r, s , it is the portion of that bidisc satisfying $c^{rs}|x|^s|y|^r < 1$. In the dependent case a primitive-ray expansion continues the series meromorphically to the whole bidisc and identifies every polar hypersurface, with noncancelling simple slice residues. In both cases no point of the bidisc boundary admits joint meromorphic continuation. The boundary proof combines a dense positive atomic measure on real parameter slices with a classical subharmonic propagation argument and, where necessary, interior pole grids. The conclusion is genuinely joint: it does not require every individual complex slice to have a natural boundary.

1 Introduction

Two commuting circle endomorphisms have common periodic points at two independent times. For multiplication by a and by b , the common-return count at (n, m) is the elementary arithmetic quantity $\gcd(a^n - 1, b^m - 1)$. We retain both times in the ordinary series

$$R_{a,b}(x, y) = \sum_{n,m \geq 1} \gcd(a^n - 1, b^m - 1) x^n y^m, \quad a, b \geq 2. \quad (1)$$

Our question is the full complex-analytic geometry of this series: where it converges absolutely, which poles appear after continuation, and which points obstruct joint meromorphic continuation.

Theorem 2.1 answers these questions for every pair of integer bases. Multiplicative dependence changes the initial convergence domain and creates an explicit locally finite polar divisor inside the bidisc. It does not change the terminal joint boundary: the entire boundary of the bidisc obstructs meromorphic continuation in both branches. In particular, the curved part of the dependent convergence boundary must not be confused with the final joint natural boundary.

Neither the dynamics nor the use of rectangular return times is new. Ward's thesis already defines a rectangular zeta function, aggregating rectangles by volume into one variable [7, Chapter IV, Definition IV.1.2]. Ward's later paper explicitly treats the two independent times of the multiplication by 2 and 3 circle action and its solenoid extension [6, Example 3.1]. These are direct precedents for the native count and the two clocks in (1). Merely keeping a second variable would not constitute the analytic classification proved here.

There are also close one-variable invariants for commuting systems. Miles studies weak and strong synchronization through equalities between corresponding iterates at a common time, together with generating functions and arithmetic growth bounds [3]. His natural-boundary theorem for

Object	Time or weight retained	Analytic object
Ward [7]	Rectangular volume	One-variable rectangular zeta
Miles [3]	One common iterate time	Synchronization generating functions
Miles [4]	All finite-index subgroups, by index	One-variable dynamical zeta
Equation (1)	Two independent rectangular times	Ordinary bivariate series; joint domain, poles and boundary

Table 1: The comparison is between different generating objects, not between different names for the same zeta function. Ward [6] already supplies the native two-clock circle example.

commuting group automorphisms concerns the dynamical zeta function summing over all finite-index subgroups by their index [4]. Rectangular subgroups form a special subfamily of those subgroups, but the ordinary series (1) is not an index-weighted specialization of that one-variable analytic problem.

The diagonal sequence has its own current classification. Nguyen-Dang proves that $\gcd(a^n - 1, b^n - 1)$ satisfies a constant-coefficient linear recurrence exactly when the bases are multiplicatively dependent [5, Theorem 1.1]. We do not claim the corresponding diagonal rationality or natural-boundary consequences as new. The simplest dependent pair already illustrates the distinction: for $a = b = 2$, the diagonal generating function is

$$\sum_{n \geq 1} (2^n - 1)z^n = \frac{2z}{1 - 2z} - \frac{z}{1 - z},$$

whereas (1) has genuine interior hypersurfaces $2x^i y^j = 1$, for every primitive positive pair (i, j) , and no joint continuation across any bidisc boundary point. A rational diagonal therefore does not recover the two-variable polar or continuation geometry.

Our only deep arithmetic input is the two-exponent S -unit gcd estimate of Corvaja–Zannier [1]. The independent convergence-domain argument is a direct deduction from that theorem, not a new gcd bound. The analytic ingredients—totient expansions, positive atomic Cauchy transforms, primitive-ray summation and subharmonic coefficient envelopes—are classical as well. We prove the precise propagation statement needed here using the standard principles described by Korevaar–Wiegerinck [2]. The work is their complete application to this return family, including actual noncancelling poles and the parameter regimes that cannot be handled by a holomorphic-product argument.

The comparison locators for [1, 3, 4, 6] refer to the original or accepted manuscripts specified in the bibliography and source record. Final publication metadata does not assert unchanged theorem numbering. The diagonal reference [5] is the public preprint version of 25 August 2026.

2 Common returns and the joint classification

Let $T_a, T_b : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}/\mathbb{Z}$ be multiplication by $a, b \geq 2$. They commute and define an $\mathbb{N}^2$ -action. For $n, m \geq 1$, put

$$C_{a,b}(n, m) = \#(\text{Fix}(T_a^n) \cap \text{Fix}(T_b^m)).$$

The common kernel of multiplication by positive integers A, B on $\mathbb{R}/\mathbb{Z}$ is the cyclic subgroup of order $\gcd(A, B)$. Indeed, if $d = \gcd(A, B)$, Bézout’s identity shows that being killed by A and B is

equivalent to being killed by d . Consequently

$$C_{a,b}(n, m) = \gcd(a^n - 1, b^m - 1). \quad (2)$$

The same counts can be realized by the automorphisms dual to multiplication by a, b on $\mathbb{Z}[(ab)^{-1}]$: the gcd in (2) is coprime to ab , so localization does not change the common finite quotient. We shall only use the circle realization.

Write $\mathbb{D} = \{|z| < 1\}$, $\mathbb{T} = \partial\mathbb{D}$, and

$$\mathcal{P} = \{(i, j) \in \mathbb{Z}_{\geq 1}^2 : \gcd(i, j) = 1\}.$$

The *open absolute-convergence domain* of a double series is the interior of the set on which the sum of absolute values is finite. This convention matters on the coordinate axes: every summand in (1) is zero if $x = 0$ or $y = 0$, even at points outside the open domain.

A *joint meromorphic cap* at a boundary point of a domain is a meromorphic function on a neighborhood of that point which agrees with the given function on the interior overlap. All continuation statements below have this local meaning.

Theorem 2.1 (Two-clock analytic classification). *Let $a, b \geq 2$ be integers.*

1. *If a, b are multiplicatively independent, the open absolute-convergence domain of $R_{a,b}$ is exactly $\mathbb{D}^2$.*
2. *If they are multiplicatively dependent, write uniquely*

$$a = c^r, \quad b = c^s, \quad c \in \mathbb{Z}_{\geq 2}, \quad r, s \geq 1, \quad \gcd(r, s) = 1.$$

Then the open absolute-convergence domain is

$$\Omega_{r,s,c} = \{(x, y) \in \mathbb{D}^2 : c^{rs}|x|^s|y|^r < 1\}. \quad (3)$$

On this domain the series agrees with

$$R_{a,b}(x, y) = \sum_{(i,j) \in \mathcal{P}} \frac{c^{h_{i,j}} x^i y^j}{1 - c^{h_{i,j}} x^i y^j} - \frac{xy}{(1-x)(1-y)}, \quad h_{i,j} = \gcd(ri, sj). \quad (4)$$

This expansion extends it meromorphically to $\mathbb{D}^2$: locally, it is a finite sum of meromorphic terms plus a normally convergent holomorphic tail. Its polar hypersurfaces there are exactly

$$P_{i,j} = \{(x, y) \in \mathbb{D}^2 : c^{h_{i,j}} x^i y^j = 1\}, \quad (i, j) \in \mathcal{P}. \quad (5)$$

Every component is genuine and has pole order one. For fixed $y_0 \in \mathbb{D}$, the residue at any slice pole $x_0 \in \mathbb{D}$ is

$$\text{Res}_{x=x_0} R_{a,b}(x, y_0) = -x_0 \sum_{\substack{(i,j) \in \mathcal{P} \\ c^{h_{i,j}} x_0^i y_0^j = 1}} \frac{1}{i} \neq 0. \quad (6)$$

The sum is finite.

3. *In both branches the resulting meromorphic function on $\mathbb{D}^2$ has no joint meromorphic cap at any point of $\partial(\mathbb{D}^2)$.*

The third assertion is joint, not slice-wise. For instance, $R_{a,b}(x, 0) = 0$ is an entire slice. Nevertheless a neighborhood of $(x_0, 0)$, $|x_0| = 1$, includes other parameter values, and the theorem excludes a joint cap there. In the dependent case, $\Omega_{r,s,c}$ is the initial power-series domain, whereas $\mathbb{D}^2$ is the larger meromorphic domain with the specified interior poles. The theorem does not label every point of the curved part of $\partial\Omega_{r,s,c}$ a singularity.

3 Arithmetic input and exact convergence domains

3.1 The common-base identities

If a, b are multiplicatively dependent, take $a^u = b^v$ with $\gcd(u, v) = 1$. For every prime p , $u v_p(a) = v v_p(b)$, so $v \mid v_p(a)$ and $u \mid v_p(b)$. Thus $a = c^v, b = c^u$ for the integer

$$c = \prod_p p^{v_p(a)/v}.$$

The coprime exponent pair and then c are unique by the same valuation comparison. The integer c need not be power-free. Set $r = v, s = u$.

For positive $u \geq v$, subtraction gives

$$(c^u - 1) - c^{u-v}(c^v - 1) = c^{u-v} - 1.$$

The Euclidean algorithm on the exponents therefore proves

$$\gcd(c^u - 1, c^v - 1) = c^{\gcd(u, v)} - 1. \quad (7)$$

In particular, the dependent coefficient is $C_{a,b}(n, m) = c^{\gcd(rn, sm)} - 1$.

3.2 Multiplicatively independent bases

We use the following consequence of the general S -unit theorem of Corvaja–Zannier [1, Corollary 1 and (1.3), public version 2]. For each finite prime set S and each $\epsilon > 0$, positive integer S -units u, v that are multiplicatively independent satisfy

$$\gcd(u - 1, v - 1) < \max(u, v)^\epsilon$$

apart from finitely many exceptional pairs. This is a statement about two independent exponents, not merely a diagonal estimate with a shared exponent.

Take S to contain the prime divisors of ab . If a, b are multiplicatively independent, so are a^n, b^m for every $n, m \geq 1$. Absorbing the finite exceptional set into a constant K_ϵ gives the uniform estimate

$$C_{a,b}(n, m) \leq K_\epsilon \exp(\epsilon \max(n \log a, m \log b)). \quad (8)$$

Let $0 < u, v < 1$ and $L = \max(\log a, \log b)$. Choose $\epsilon > 0$ such that $ue^{\epsilon L} < 1$ and $ve^{\epsilon L} < 1$. Since the maximum in (8) is at most $L(n + m)$, the double series is bounded on $|x| \leq u, |y| \leq v$ by a constant times

$$\sum_{n, m \geq 1} (ue^{\epsilon L})^n (ve^{\epsilon L})^m < \infty.$$

This proves locally normal absolute convergence on $\mathbb{D}^2$.

For $y \neq 0$ and $|x| \geq 1$, a fixed row $m \geq 1$ has divergent absolute sum, since $C_{a,b}(n, m) \geq 1$. The symmetric assertion holds for $x \neq 0, |y| \geq 1$. Although the series vanishes pointwise on either coordinate axis, an axis point outside $\mathbb{D}^2$ has no neighborhood of absolute convergence. This proves the first part of Theorem 2.1.

3.3 Multiplicatively dependent bases

Put $u = |x|, v = |y|$, initially with $0 < u, v < 1$. The inequality in (3) is equivalent to $cu^{1/r}v^{1/s} < 1$. Choose $0 < \theta < 1$ sufficiently close to one that $cu^{\theta/r}v^{\theta/s} < 1$. For $g = \gcd(rn, sm)$, the bounds $n \geq g/r$ and $m \geq g/s$ imply

$$\begin{aligned} C_{a,b}(n, m)u^n v^m &\leq c^g u^n v^m \\ &\leq (cu^{\theta/r}v^{\theta/s})^g u^{(1-\theta)n} v^{(1-\theta)m} \\ &\leq u^{(1-\theta)n} v^{(1-\theta)m}. \end{aligned}$$

The last expression is summable. The strict inequalities allow the same estimate on a slightly larger closed polydisc around each nonaxis point of $\Omega_{r,s,c}$. At an axis point in this domain, choose a small positive radius for the zero coordinate and a radius less than one for the other so that the same strict inequality holds. Thus the series is locally normally convergent throughout $\Omega_{r,s,c}$.

Conversely, if $c^{rs}u^s v^r \geq 1$, the terms on the distinguished primitive ray $(n, m) = (sk, rk)$ are

$$(c^{rsk} - 1)u^{sk}v^{rk},$$

and they do not tend to zero. If instead $u \geq 1$ or $v \geq 1$ with both coordinates nonzero, the fixed-row argument applies. Together with the axis convention, these facts give exactly (3) as the open absolute-convergence domain. They establish convergence geometry; they do not preclude meromorphic passage through parts of its boundary.

4 Primitive rays and the full interior polar divisor

Assume $a = c^r, b = c^s$ with $\gcd(r, s) = 1$. Every positive lattice point has a unique decomposition $(n, m) = k(i, j)$, $k \geq 1$, $(i, j) \in \mathcal{P}$. With $h_{i,j} = \gcd(ri, sj)$, summation on one ray gives

$$\sum_{k \geq 1} (c^{kh_{i,j}} - 1)x^{ki}y^{kj} = \frac{c^{h_{i,j}}x^i y^j}{1 - c^{h_{i,j}}x^i y^j} - \frac{x^i y^j}{1 - x^i y^j}.$$

Absolute convergence on $\Omega_{r,s,c}$ justifies rearrangement. Moreover,

$$\sum_{(i,j) \in \mathcal{P}} \frac{x^i y^j}{1 - x^i y^j} = \sum_{n,m \geq 1} x^n y^m = \frac{xy}{(1-x)(1-y)}.$$

This proves (4) on the original domain.

Lemma 4.1. *For every $(i, j) \in \mathcal{P}$,*

$$h_{i,j} = \gcd(r, j) \gcd(s, i) \mid rs. \quad (9)$$

Proof. Fix a prime ℓ . Write $\alpha = v_\ell(r), \beta = v_\ell(s)$ and $\gamma = v_\ell(i), \delta = v_\ell(j)$. The coprimality assumptions give $\min(\alpha, \beta) = \min(\gamma, \delta) = 0$. If $\alpha > 0$, then $\beta = 0$, and $\min(\alpha + \gamma, \delta) = \min(\alpha, \delta)$, since either $\gamma = 0$ or $\delta = 0$. The case $\beta > 0$ is symmetric, and if $\alpha = \beta = 0$, the minimum is zero. Hence the valuation of $h_{i,j}$ equals that of the right-hand side of (9), proving both assertions. $\square$

On a compact sub-bidisc $|x| \leq u < 1, |y| \leq v < 1$, the numerator monomials satisfy

$$|c^{h_{i,j}}x^i y^j| \leq c^{rs}u^i v^j.$$

Only finitely many pairs have the right-hand side greater than $1/2$. For all remaining pairs, the corresponding term in (4) is holomorphic and bounded by $2c^{rs}u^i v^j$. This is a summable family. Thus the expansion is locally a finite meromorphic sum plus a normally convergent holomorphic tail. It defines a meromorphic function on all of $\mathbb{D}^2$, with a locally finite family of possible polar hypersurfaces $P_{i,j}$.

We now check that these are actual distinct components. For a primitive vector (i, j) , choose a second integer vector so that the two form a unimodular matrix. The resulting invertible monomial coordinate change on $(\mathbb{C}^\times)^2$ makes $x^i y^j$ one coordinate. Hence its nonzero level sets are smooth and irreducible in the algebraic torus. The portion of a level set in $\mathbb{D}^2$ is connected: after choosing a point (α, β) on it, it has the parametrization

$$x = \alpha t^j, \quad y = \beta t^{-i}, \quad t \in \mathbb{C}^\times,$$

and the bidisc inequalities specify an annulus in t . The portion is nonempty, for example by taking positive $x = c^{-h_{i,j}/(2i)}$ and $y = c^{-h_{i,j}/(2j)}$. Distinct primitive exponent vectors give distinct hypersurfaces.

At a point of one component away from all the others, its own summand has a simple pole and all other summands are holomorphic. Such points exist because the family is locally finite and distinct components have discrete intersections on each component. Thus every component is genuine and has divisor order one; intersections do not erase any part of the polar divisor.

There is also a direct slice calculation that rules out cancellation at an intersection. Fix $y_0 \in \mathbb{D}$ and a pole $x_0 \in \mathbb{D}$. Neither coordinate is zero. For each summand with $c^{h_{i,j}} x_0^i y_0^j = 1$, differentiation of its denominator gives

$$\text{Res}_{x=x_0} \frac{c^{h_{i,j}} x^i y_0^j}{1 - c^{h_{i,j}} x^i y_0^j} = -\frac{x_0}{i}.$$

Only finitely many such summands meet a compact neighborhood of (x_0, y_0) , and the remaining terms are holomorphic there. Their residues sum to (6), which is nonzero because $x_0 \neq 0$ and every $1/i$ is positive. This completes the continuation and polar-divisor assertions of Theorem 2.1.

5 Dense atomic boundaries on positive real slices

The following argument works for every pair of bases, regardless of multiplicative dependence.

Proposition 5.1. *For every real t with $0 < t < 1/b$, the function $x \mapsto R_{a,b}(x, t)$ is holomorphic in $\mathbb{D}$ and has the whole unit circle as a natural boundary, even for meromorphic continuation.*

Proof. The elementary estimate $C_{a,b}(n, m) \leq b^m - 1$ shows that

$$R_{a,b}(x, t) = \sum_{n \geq 1} A_n(t) x^n, \quad A_n(t) = \sum_{m \geq 1} C_{a,b}(n, m) t^m$$

has coefficients bounded uniformly in n . For d coprime to a , put $k_a(d) = \text{ord}_d(a)$, with $k_a(1) = 1$. Let ν_k be uniform probability on the k -th roots of unity. The totient identity $N = \sum_{d|N} \varphi(d)$ gives

$$C_{a,b}(n, m) = \sum_{\substack{d|b^m-1 \\ (d,a)=1}} \varphi(d) \mathbf{1}_{k_a(d)|n}. \quad (10)$$

Indeed, the divisors common to $a^n - 1$ and $b^m - 1$ are precisely those in this sum for which $k_a(d) \mid n$.

Define a positive atomic measure on $\mathbb{T}$ by

$$\mu_t = \sum_{m \geq 1} t^m \sum_{\substack{d | b^m - 1 \\ (d, a) = 1}} \varphi(d) \nu_{k_a(d)}. \quad (11)$$

Its mass is finite:

$$\mu_t(\mathbb{T}) \leq \sum_{m \geq 1} (b^m - 1) t^m < \infty.$$

Character orthogonality on a finite cyclic group gives $\int \xi^n d\nu_k(\xi) = \mathbf{1}_{k|n}$. Consequently (10) implies

$$A_n(t) = \int_{\mathbb{T}} \xi^n d\mu_t(\xi).$$

Choose a prime $\ell \nmid ab$. For every $j \geq 1$, $\ell^j \mid b^{m_j} - 1$ when $m_j = \text{ord}_{\ell^j}(b)$. Thus μ_t contains a positive multiple of ν_{k_j} , where $k_j = \text{ord}_{\ell^j}(a)$. The integers k_j are unbounded. Otherwise, if they were bounded by K , their orders would all divide $N = \text{lcm}(1, \dots, K)$; hence $\ell^j \mid a^N - 1$ for every j , impossible for $a \geq 2$. The union of the complete k_j -th root grids is therefore dense in $\mathbb{T}$. Positivity in (11) makes every displayed atom a nonzero mass of the actual measure.

For $|x| < 1$, finite total mass permits the geometric series to be integrated:

$$R_{a,b}(x, t) = \int_{\mathbb{T}} \frac{\xi x}{1 - \xi x} d\mu_t(\xi).$$

For $\xi, \xi_0 \in \mathbb{T}$ and $0 < \varrho < 1$,

$$\left| (1 - \varrho) \frac{\varrho \xi \xi_0^{-1}}{1 - \varrho \xi \xi_0^{-1}} \right| \leq 1.$$

The integrand tends to the indicator of $\xi = \xi_0$. Dominated convergence gives

$$\lim_{\varrho \uparrow 1} (1 - \varrho) R_{a,b}(\varrho \xi_0^{-1}, t) = \mu_t(\{\xi_0\}). \quad (12)$$

At every nonzero atom this limit precludes a holomorphic extension at ξ_0^{-1} . Those points are dense in the unit circle. A meromorphic continuation across an arc would require a pole at each of them, contradicting discreteness of poles. This proves the proposition. $\square$

No quantitative gcd estimate is used in this slice argument. The positive real parameter is what ensures noncancellation in the measure; no analogous positivity assertion is made for arbitrary complex parameter values.

6 Propagation and the joint meromorphic boundary

6.1 A classical subharmonic propagation argument

We give the precise local statement needed to pass from Proposition 5.1 to a joint boundary. Its ingredients are the standard upper-envelope and decreasing-limit principles for subharmonic functions, Cauchy's inequality and the maximum principle; see [2, Section 4.8 and Properties 8.4.3]. The lemma is an explicit application of these classical tools.

Lemma 6.1 (Joint-cap propagation). *Let $V \subset \mathbb{C}$ be a connected domain and $F \in \mathcal{O}(\mathbb{D} \times V)$. Suppose that for one $t \in V$, the slice $F(\cdot, t)$ has the whole unit circle as a holomorphic natural boundary. Then F has no local joint holomorphic or meromorphic cap through any point of $\mathbb{T} \times V$.*

Proof. Suppose first that there is a holomorphic cap H on $B(x_0, \rho) \times B(y_0, \eta)$, where $|x_0| = 1$ and $\overline{B(y_0, \eta)} \subset V$. It agrees with F on the interior overlap; shrinking the product if necessary, the identity theorem gives agreement throughout that connected overlap. Choose

$$0 < \delta < \min(1/2, \rho/4), \quad z_c = (1 - \delta)x_0, \quad M = -\log \delta.$$

For $n \geq 0$, the Taylor coefficients

$$a_n(y) = \frac{1}{n!} \partial_x^n F(z_c, y)$$

are holomorphic on V . For $n \geq 1$, let $u_n(y) = n^{-1} \log |a_n(y)|$, with $\log 0 = -\infty$. Each u_n is subharmonic, allowing the identically negative-infinite function.

For any $0 < q < \delta$ and compact $K \subset V$, Cauchy's inequality on the circle $|x - z_c| = q$, uniformly for $y \in K$, gives a constant $C_{q,K} \geq 1$ such that

$$u_n(y) \leq -\log q + \frac{\log C_{q,K}}{n} \quad (y \in K, n \geq 1). \quad (13)$$

Thus the family is locally uniformly bounded above. For a locally upper-bounded function h , denote its upper semicontinuous regularization by

$$h^*(y) = \lim_{\varepsilon \downarrow 0} \sup_{|w-y| < \varepsilon} h(w).$$

Define the regularized tail envelopes

$$V_N = \left(\sup_{n \geq N} u_n \right)^*, \quad V_\infty = \lim_{N \rightarrow \infty} V_N. \quad (14)$$

The upper-envelope theorem makes each V_N subharmonic. They decrease with N , so V_∞ is subharmonic or identically $-\infty$.

Apply (13) on a compact neighborhood of an arbitrary parameter point before regularizing. On its interior this gives

$$V_N \leq -\log q + \frac{\log C_{q,K}}{N}.$$

Letting $N \rightarrow \infty$ and then $q \uparrow \delta$, we obtain

$$V_\infty(y) \leq M \quad (y \in V). \quad (15)$$

Regularization only raises the supremum; consequently

$$V_\infty(t) \geq \limsup_{n \rightarrow \infty} u_n(t).$$

The Taylor series of $F(\cdot, t)$ about z_c has radius exactly δ . It has at least that radius since $B(z_c, \delta) \subset \mathbb{D}$; a larger radius would continue the slice through x_0 , contrary to its natural boundary. Cauchy–Hadamard therefore gives $\limsup_n u_n(t) = -\log \delta = M$. Together with (15), this yields $V_\infty(t) = M$. The strong maximum principle on connected V forces $V_\infty \equiv M$.

But $\overline{B(z_c, 2\delta)} \subset B(x_0, \rho)$, because $|x - x_0| \leq 2\delta + \delta < \rho$ there. For a smaller parameter disc $B(y_0, \eta')$, the cap and Cauchy's inequality now give

$$|a_n(y)| \leq C'(2\delta)^{-n} \quad (y \in \overline{B(y_0, \eta')}).$$

These are the same coefficients, since $H = F$ near z_c . Applying the tail-envelope argument on the interior of this disc yields $V_\infty \leq -\log(2\delta) < M$, a contradiction. This excludes holomorphic caps.

For a meromorphic cap, shrink to a product where it can be written as A/B , with A, B holomorphic and $B \not\equiv 0$. The denominator cannot vanish on an entire product of a unit-circle arc and an open parameter disc. If it did, for every fixed parameter the one-variable identity theorem in x would make B identically zero on the first disc, and thus zero on the whole product. There is therefore a nearby face point at which $B \neq 0$. In a smaller neighborhood it supplies a holomorphic cap, already excluded. This proves the meromorphic assertion. $\square$

The regularized tails in (14) avoid any assertion that a raw pointwise logarithmic limsup is subharmonic, or that it equals its regularization at every parameter. The distinction is essential for exceptional slices: if $h \in \mathcal{O}(\mathbb{D})$ has a natural boundary, then $F(x, y) = (y - y_0)h(x)$ has an entire zero slice at y_0 , yet still has no joint cap anywhere on $\mathbb{T} \times V$.

6.2 The independent branch

For multiplicatively independent bases, Section 3 proves holomorphy of $R_{a,b}$ on $\mathbb{D}^2$. Choose $t \in (0, 1/b)$, apply Proposition 5.1, and then apply Lemma 6.1 with $V = \mathbb{D}$. This excludes every joint meromorphic cap on $\mathbb{T} \times \mathbb{D}$. Since $R_{a,b}(x, y) = R_{b,a}(y, x)$, the same argument excludes $\mathbb{D} \times \mathbb{T}$. A cap at a corner in $\mathbb{T}^2$ would contain a nearby point of one of these faces and restrict to a cap there. Thus every point of $\partial(\mathbb{D}^2)$ is excluded in this branch.

6.3 The dependent branch and its parameter threshold

The dependent continuation has genuine poles in the bidisc, so one must not apply Lemma 6.1 to it on all of $\mathbb{D}^2$. The coefficient bound $C_{a,b}(n, m) \leq b^m - 1$ does, however, give joint holomorphy on

$$\mathbb{D} \times \mathbb{D}_{1/b}, \quad \mathbb{D}_{1/b} = \{|y| < 1/b\}.$$

Proposition 5.1 and Lemma 6.1, now with $V = \mathbb{D}_{1/b}$, exclude every joint meromorphic cap at a point (x_0, y_0) with $|x_0| = 1$ and $|y_0| < 1/b$.

For $1/b < |y_0| < 1$, we instead use the actual poles already identified in Section 4. Take arbitrarily large integers k with $\gcd(k, r) = 1$. The pair (sk, r) is primitive, and

$$h_{sk,r} = \gcd(rsk, sr) = rs.$$

The slice poles include all the roots of

$$c^{rs} x^{sk} y_0^r = 1. \tag{16}$$

Their modulus is

$$(c^{rs} |y_0|^r)^{-1/(sk)} < 1,$$

tending to one, and their complete root grids have mesh tending to zero. Formula (6) proves that every such pole is genuine, even if other components meet there. Near each $x_0 \in \mathbb{T}$, infinitely many interior slice poles therefore tend to x_0 .

Suppose a joint meromorphic cap A/B existed at such a (x_0, y_0) . Shrink its parameter disc so that $1/b < |y| < 1$ throughout it. For *each* parameter y in that disc, (16) gives infinitely many genuine slice poles in the cap's x -disc, tending to x_0 . The denominator must vanish at each: otherwise A/B would be holomorphic there and could not agree with the continued series. The

one-variable identity theorem makes $B(\cdot, y)$ identically zero on the x -disc. This for every y in an open disc contradicts $B \not\equiv 0$.

If $|y_0| = 1/b$, any putative cap contains nearby face points with $|y| > 1/b$, just excluded. The strip and pole arguments together therefore exclude all of $\mathbb{T} \times \mathbb{D}$. Symmetry excludes the other face, and the same corner argument as above excludes $\mathbb{T}^2$. This completes the proof of Theorem 2.1.

7 Consequences and limits

For $a = b = 2$, the initial domain is

$$\{(x, y) \in \mathbb{D}^2 : 2|xy| < 1\},$$

and the continued function has the polar components $2x^i y^j = 1$ for $(i, j) \in \mathcal{P}$. This specialization illustrates both distinctions in the theorem: the diagonal series is rational, but the joint bidisc boundary is impenetrable; and the curved boundary of initial convergence is not everywhere a barrier to continuation. For example, $(x, y) = (i/\sqrt{2}, i/\sqrt{2})$ lies on $2|xy| = 1$ but on no polar component. The pair $(i, j) = (1, 1)$ gives $2xy = -1$, while every other positive primitive pair has $i + j > 2$ and hence $|2x^i y^j| < 1$. The continued function is therefore holomorphic near this point. This is a direct illustration of the classified divisor, not an additional independent result.

The arithmetic and analytic inputs have separate roles. Corvaja–Zannier supplies the subexponential two-index bound in the independent branch. Elementary common-base algebra determines the dependent convergence domain and ray exponents. Positive atomic measures give one family of natural-boundary slices for both branches. Classical subharmonic propagation turns those slices into joint boundary obstructions where a holomorphic product is available, and the actual dependent pole grids settle the remaining parameter range.

The result does not assert that every fixed complex slice has a natural boundary; the zero axes show why that would be false. Nor does it give an effective exceptional threshold for the S -unit theorem, a new diagonal recurrence classification, or a new definition of rectangular returns. It concerns the ordinary two-clock series (1) and its native common fixed counts, without replacing the two integer times by volume, subgroup index, or another arithmetic clock.

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